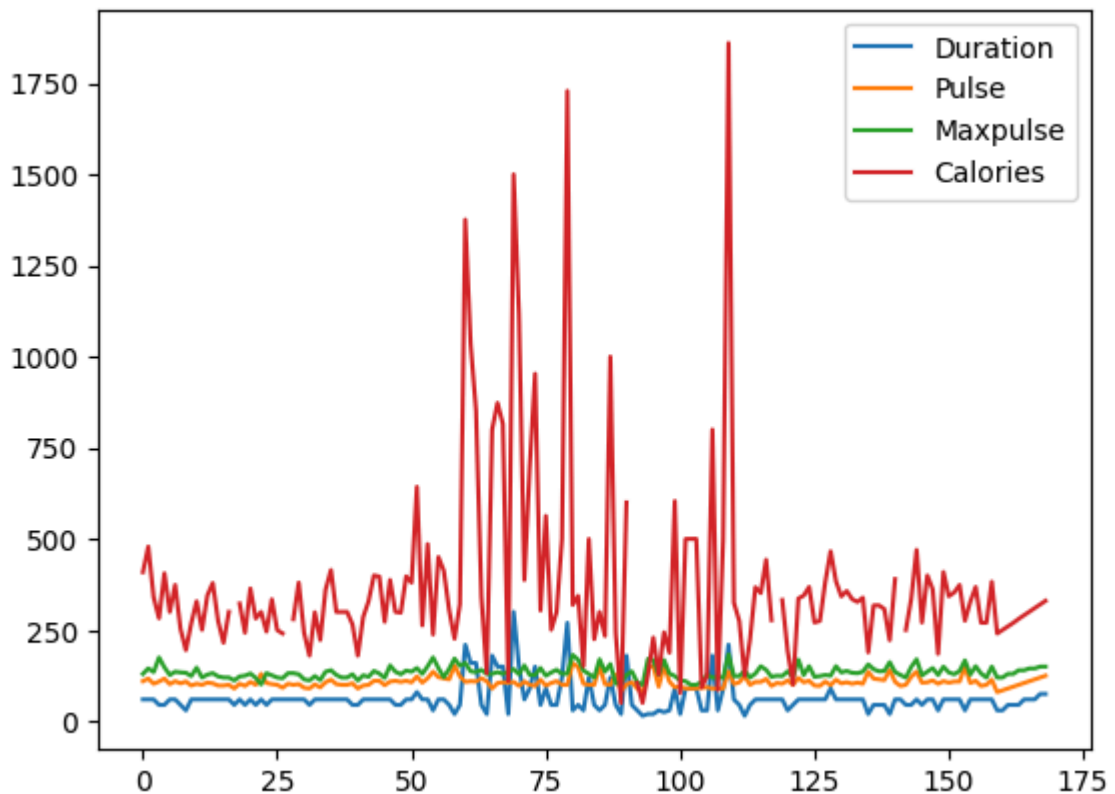


Pandas - Plotting

1. Plotting

Pandas uses the `plot()` method to create diagrams. We can use Pyplot, a submodule of the Matplotlib library to visualize the diagram on the screen.

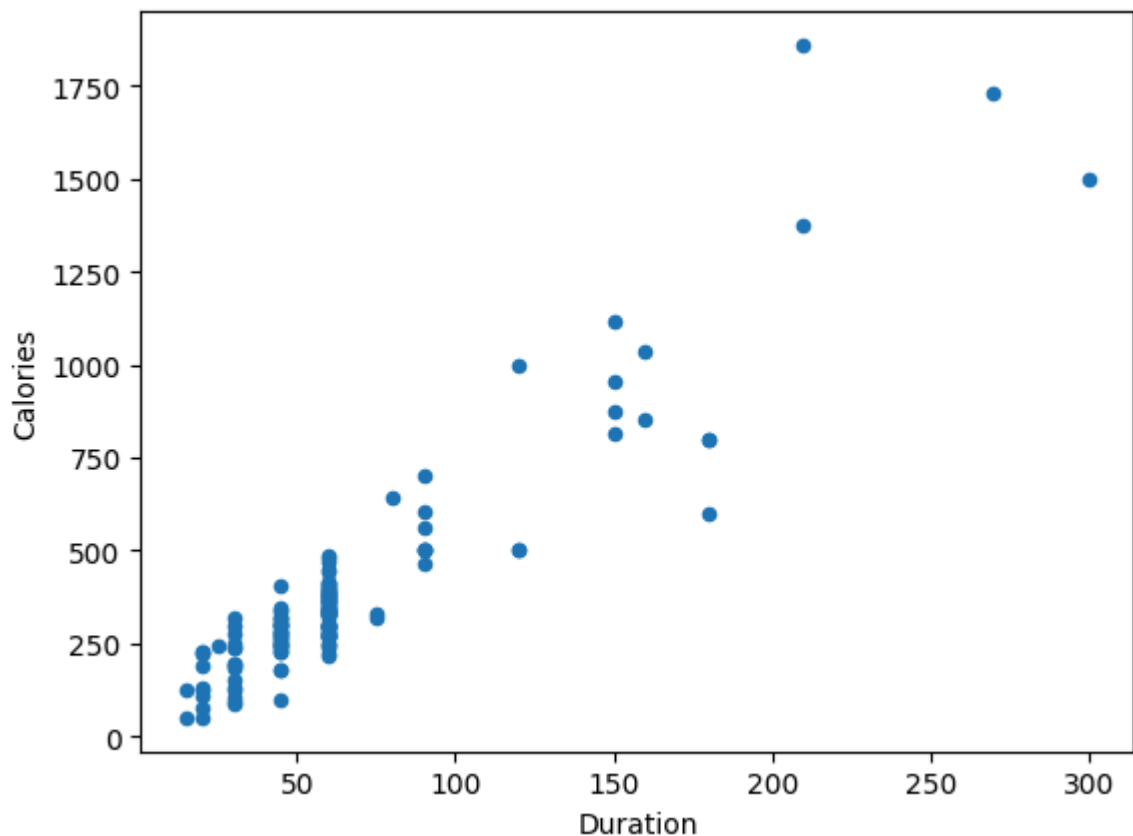
```
In [1]: #Import pyplot from Matplotlib and visualize our DataFrame:  
import pandas as pd  
import matplotlib.pyplot as plt  
df = pd.read_csv('data/data.csv')  
df.plot()  
plt.show()
```



2. Scatter Plot

Specify that you want a scatter plot with the `kind` argument: `kind = 'scatter'`. A scatter plot needs an x- and a y-axis. In the example below we will use "Duration" for the x-axis and "Calories" for the y-axis. Include the x and y arguments like this: `x = 'Duration', y = 'Calories'`

```
In [3]: import pandas as pd  
import matplotlib.pyplot as plt  
df = pd.read_csv('data/data.csv')  
df.plot(kind = 'scatter', x = 'Duration', y = 'Calories')  
plt.show()
```

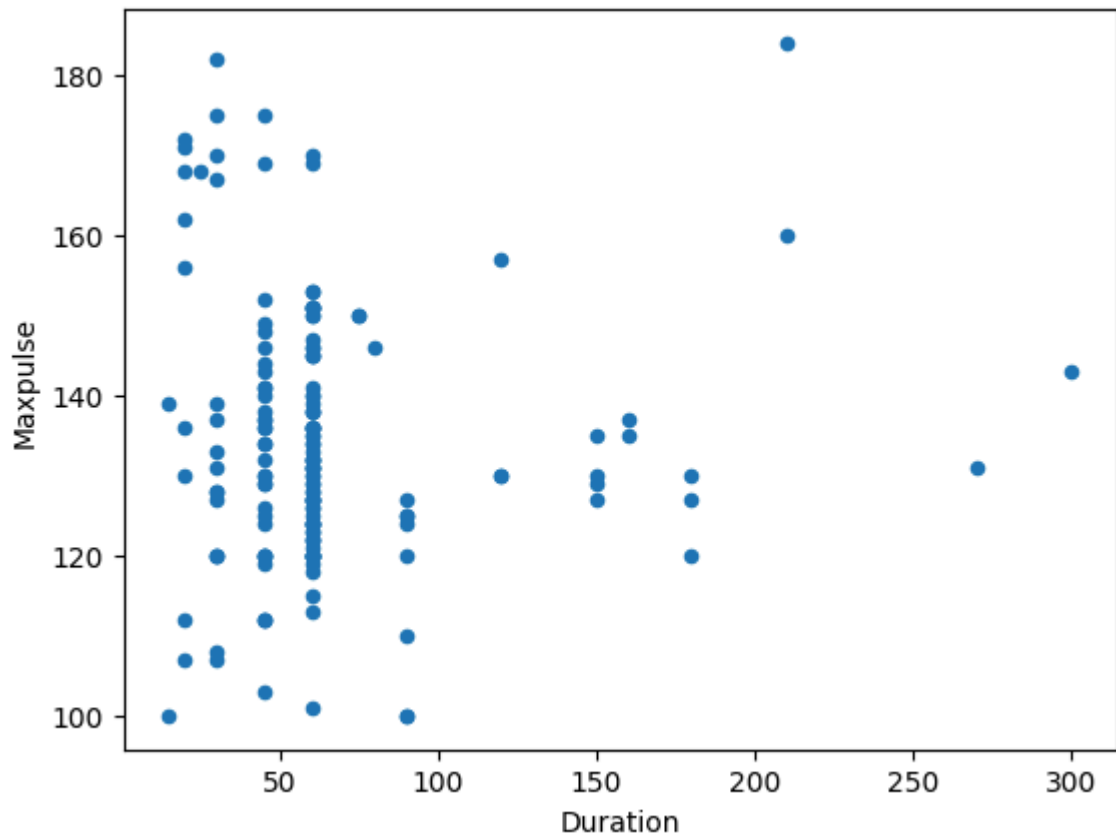


Remember: In the previous example, we learned that the correlation between "Duration" and "Calories" was 0.922721, and we concluded with the fact that higher duration means more calories burned.

By looking at the scatterplot, I will agree.

Let's create another scatterplot, where there is a bad relationship between the columns, like "Duration" and "Maxpulse", with the correlation 0.009403:

```
In [4]: # A scatterplot where there are no relationship between the columns:
import pandas as pd
import matplotlib.pyplot as plt
df = pd.read_csv('data/data.csv')
df.plot(kind = 'scatter', x = 'Duration', y = 'Maxpulse')
plt.show()
```

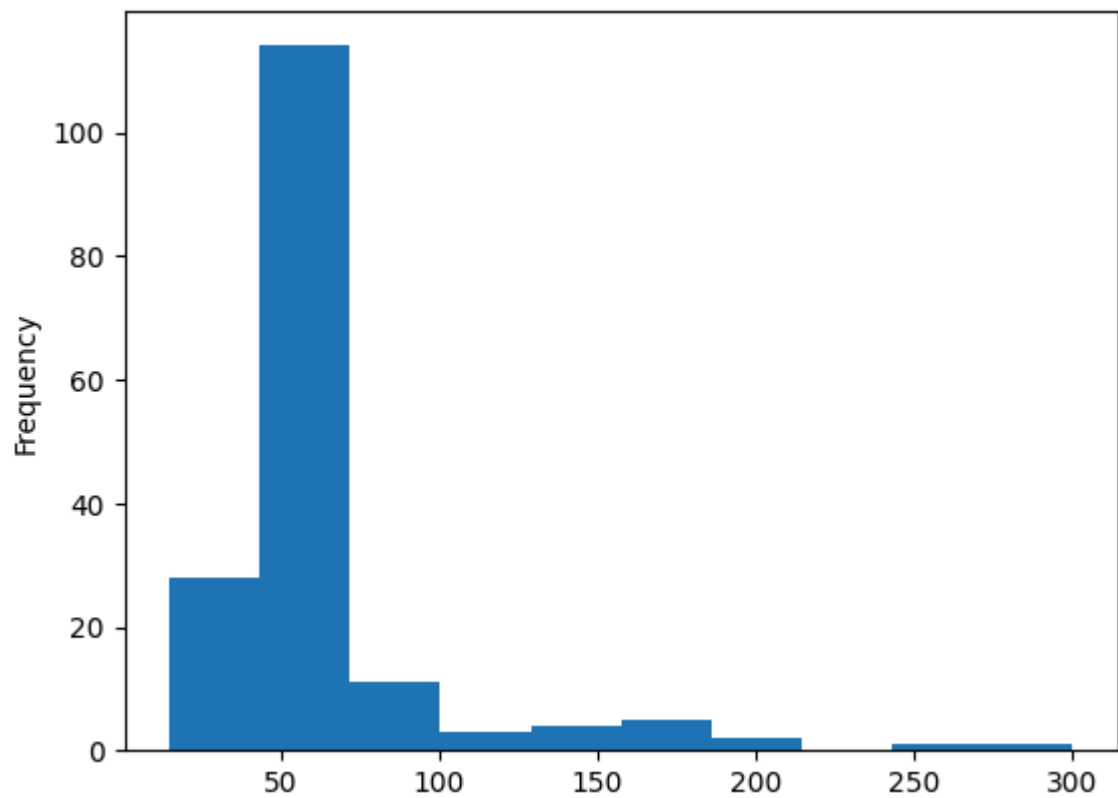


3. Histogram

Use the `kind` argument to specify that you want a histogram: `kind = 'hist'`. A histogram needs only one column. A histogram shows us the frequency of each interval, e.g. how many workouts lasted between 50 and 60 minutes? In the example below we will use the "Duration" column to create the histogram:

```
In [6]: df["Duration"].plot(kind = 'hist')
```

```
Out[6]: <Axes: ylabel='Frequency'>
```



Note: The histogram tells us that there were over 100 workouts that lasted between 50 and 60 minutes.